

ISYS90090
Cybersecurity
Management

MODULE 3
ASSIGNMENT 2
ATIF AHMAD



The background of the image is a piece of marbled paper with a complex, organic pattern. It features swirling, cell-like shapes in various shades of brown, from light tan and beige to deep, dark chocolate and near-black tones. There are occasional streaks and spots of a muted teal or blue-green color interspersed within the brown pattern. The overall effect is reminiscent of traditional marbling techniques used in bookbinding.

**If you think technology can
solve your security problems, then
you don't understand the problems
and you don't understand the
technology.**

Mocak
Bruce Schneier

The background of the slide is a dark, moody image of a stormy sky. A bright, jagged lightning bolt is visible in the lower-left quadrant, extending towards the bottom right. The overall color palette is dark purple and blue.

**Security is a process,
not a product.**

Mocak

Bruce Schneier

This Week in Cybersecurity Management



- ▶ Step 1: Study Canvas Module 3
- ▶ Step 2: Same as last week's reading - prescribed textbook pages 34 to 74 on "Need for Information Security"
- ▶ Optional Step 3: Answer discussion questions on Canvas, and chat with peers
- ▶ Step 4: Attend workshop
- ▶ Optional Step 5: Continue discussion with peers on Canvas

This week's workshop



- ▶ **Part A:** Quiz
- ▶ **Part B:** Q&A session on this week's content
- ▶ **Part C:** Teaching case: Horizon Scenes 1 to 6 (maybe 1 to 3)

Intended Learning Objectives

- Identify a range of information security threats actors
- Identify common cybersecurity attacks on organisations



What are information security threats?

- ▶ Define a threat
- ▶ How do threats act in an organization?
- ▶ How do organisations mitigate the risks associated with the threat(s)?



Categories of threat

Category of threat	Examples
Compromise to intellectual property	Piracy, copyright infringement
Software attacks	Viruses, worms, macros, denial of service
Deviation in quality of service	ISP, power or WAN service issues from service providers
Espionage or trespass	Unauthorised access and/or data collection
Forces of nature	Fire, flood, earthquake, lightning
Human error or failure	Accidents, employee mistakes
Information extortion	Blackmail, information disclosure
Missing, inadequate, or incomplete organisational policy or planning	Loss of access to information systems due to disk drive failure without proper backup and recovery plan, organisational policy or planning in place
Missing, inadequate, or incomplete controls	Network compromised because of no firewall security controls
Sabotage or vandalism	Destruction of systems or information
Theft	Illegal confiscation of equipment or information
Technical hardware failures or errors	Equipment failure
Technical software failures or errors	Bugs, code problems, unknown loopholes
Technological obsolescence	Antiquated or outdated technologies

Threats to information security. Recreated from: Whitman, ME & Mattord, HJ 2012, *Principles of information security*, Boston, MA, USA, Table 2-1, p. 44.

What are information security attacks?

An attack is an action whereby a threat agent exploits a vulnerability in a system to create an impact. Note the distinction between a threat and a threat agent. Whereas the threat is a constant danger, the threat agent is the actor that creates the impact.

Vector	Description
IP scan and attack	The infected system scans a random or local range of IP addresses and targets any of several vulnerabilities known to hackers or left over from previous exploits, such as Code Red, Back Orifice, or PoizonBox.
Web browsing	If the infected system has write access to any web pages, it makes all web content files (.html, .asp, .cgi, and others) infectious, so that users who browse to those pages become infected.
Virus	Each infected machine infects certain common executable or script files on all computers to which it can write with virus code that can cause an infection.
Mass mail	By sending e-mail infections to addresses found in the address book, the infected machine infects many users, whose mail-reading programs also automatically run the programs and infect other systems.
Simple network management protocol (SNMP)	By using the widely known and common passwords that were employed in early versions of this protocol (which is used for remote management of network and computer devices), the attacking program can gain control of the device. Most vendors have closed these vulnerabilities with software updates.

Attack Replication Vectors, recreated from:: Whitman, ME & Mattord, HJ 2012, *Principles of information security*, Boston, MA, USA, Table 2-2, p.66.

- Threat:** The possibility of someone breaking into your house.
- Threat Actor:** The burglar who actually carries out the break-in.
- Threat Vector:** The open window or unlocked door they use to get inside.
- Attack:** The burglar successfully entering your house through the open window.





Horizon Teaching Case

SCENES 1 TO 6



Workshop: Horizon teaching case



- ▶ In this subject we will be following the fictional (albeit realistic) unfolding of a major information security incident at Horizon, a hypercar manufacturer. The purpose of the case study is to help ground the knowledge presented to you in a real world scenario. While reading the first six scenes of the case study consider the following bullet points:
 - The concept of 'knowledge leakage' in a firm where knowledge drives competitive advantage when reading scenes 1 to 3.
 - The distinction between knowledge and information in scene 4. The parallels between the incident at Horizon and the incident at T-mobile.
 - The range of threats and attacks when reading scenes 5 and 6.

Let's talk about Assignment 2

Q&A

